Prizma-Seq: A Parameter-Free Quadratic Delta-State Sequence Mixer

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Abstract

We present *Prizma-Seq*, a Gated-DeltaNet–family sequence mixer whose only added lever is a *parameter-free quadratic feature map* (QUAD2) that lifts each head's keys and queries from dimension d_h=32 to d_ϕ=256 using fixed seeded monomial buffers. Because the monomial coefficients are seeded constants rather than learned weights, QUAD2 adds *zero* parameters while making the per-head carried associative state *rectangular* ($d_h \times d_\phi$ rather than $d_h \times d_h$), which preserves exact $O(1)$-per-step, constant-memory inference. Parameter-matched against a tuned decoder-only Transformer (RMSNorm + SwiGLU + RoPE), Prizma-Seq clears a pre-registered, falsifiable diagnostic bar at small scale: it reaches descriptive parity on MQAR (D=128) at ∼860K matched parameters, solves D=128 at 130K parameters where the matched Transformer fails and the smallest Transformer solver on our coarse grid needs ≥461K (≥3.5× parameter-efficiency); it solves induction (0.9995) and selective-copy (0.9991); and it is competitive on a text8 character LM (1.7496 vs. 1.7254 bits-per-character, within a pre-registered +0.05 margin — it does *not* beat the Transformer). At inference its state is a constant 17.9 MB at every sequence length (28–455× less than a growing KV-cache), and a measured $O(1)$-latency crossover appears at n≥32k (2.4–2.8× faster at n=65k); below ∼16k it is ∼1.3–1.6× *slower*. A causal ablation shows the gain is the quadratic monomials specifically (QUAD2 ≫ rand_linear ≈ none ≫ TF), not "a bigger recurrent state." We make no per-FLOP, large-scale-LM, or backprop-free-parity claim; Prizma-Seq is a *candidate* efficient-attention replacement in the tested regime, and the QUAD2 kernel itself belongs to the Based/Hedgehog feature-map family — the contribution here is the *rectangular delta-state* framing.

1 Introduction

Self-attention [19]'s score matrix costs $O(n^2)$ compute and an $O(n)$ growing KV-cache at decode time, which dominates the cost of long-context generation. A now-large family of *efficient attention replacements* — linear attention [9], state-space models [8, 7], and delta-rule recurrences [21, 20] — replace the score matrix with a carried associative state of fixed size, trading the $O(n^2)$ matrix for $O(n \cdot d_h^2)$ training and, crucially, $O(d_h^2)$ *per-step, sequence-length-independent* inference. The practical promise is constant-memory, constant-latency decoding at long context.

The cost of this trade is *recall*: a fixed-rank state can store fewer associations than an unbounded KV-cache, and this shows up directly on associative-recall diagnostics such as MQAR [1]. The Based and Hedgehog lines [2, 23] recover much of this recall by passing keys and queries through an expressive (often quadratic) *feature map* before forming the state, enlarging the effective key space.

This paper studies a single, deliberately narrow question: *can a parameter-free quadratic feature map, used to make a Gated-DeltaNet state rectangular, match a tuned Transformer on the field-standard attention-diagnostic suite at matched parameters while preserving $O(1)$ inference?* Our answer, scoped to small scale and the named tasks, is yes. We call the resulting mixer *Prizma-Seq*. Its lever, QUAD2, lifts each head's keys/queries from d_h=32 to d_ϕ=256 using *fixed seeded monomial buffers*: the lift is a function with no learnable parameters, so it adds zero to the parameter budget, and it turns the per-head state from a square $d_h \times d_h$ matrix into a *rectangular* $d_h \times d_\phi$ matrix. Because the state size is still independent of n, the constant-memory / $O(1)$-per-step inference property is preserved exactly.

We test Prizma-Seq against a pre-registered, falsifiable bar (Section 3) and report every number with its honest caveat (Section 7). We emphasize up front what is *not* claimed: Prizma-Seq does not beat the Transformer on the character LM (it is within a margin), the latency advantage is long-context-only, the architecture is ∼5× slower to train per step, we make no per-FLOP claim, and we make no large-scale-LM or backprop-free-parity claim. The contribution is the **rectangular delta-state framing** of a known kernel, not a new kernel.

2 Method

2.1 The Gated-DeltaNet backbone

Prizma-Seq replaces self-attention's token mixer; everything else — the SwiGLU FFN [16], RMSNorm [22], RoPE [17] on the encoder projections, token embeddings, and the tied output head, in the Llama-family style [18] — is byte-identical to our Transformer baseline, so *only the mixer differs*. This makes the comparison a clean attention-replacement test rather than a whole-architecture confound.

Within each head, the mixer carries an associative state S_t updated by a precision-gated *delta rule* [21, 15] with a data-dependent forget gate α_t [20, 7]:

$$S_t \;=\; \alpha_t\, S_{t-1} \;+\; \beta_t\, \varepsilon_t\, k_t^\top, \qquad \varepsilon_t \;=\; v_t - S_{t-1} k_t, \tag{1}$$

where k_t are L_2-normalized keys, v_t values, $\beta_t = \sigma(W_\beta x_t)$ an input-dependent write gate, and ε_t the prediction error (the "surprise") of the current value under the running state. Reads are recognition-by-reconstruction $o_t = S_{t-1} q_t$, augmented by a small exact local-window head (w=16) of the kind used in Based/Griffin [2, 5]. A short causal depthwise convolution (kernel 4), standard in Mamba/Based/DeltaNet, precedes the projections.

Equation (1) is exactly one gradient step on a per-token *free energy* $F_t(S) = \frac{1}{2}\|v_t - S k_t\|^2$, since $\partial F_t/\partial S = -(v_t - S k_t) k_t^\top = -\varepsilon_t k_t^\top$. This predictive-coding reading [14, 6] of the delta write is the project's framing of the mechanism; the components themselves are borrowed and known-good, and we disclose this explicitly (Section 8). The distinct, separately-tested lever is the feature map below.

2.2 The quad2 parameter-free quadratic feature map

The Based/Hedgehog insight [2, 23] is that passing keys/queries through a quadratic feature map $\phi(\cdot)$ before forming the state recovers much of the recall that a linear state loses. We adopt this kernel family but in a specific, parameter-free form. For a head dimension d_h=32, define $\phi : \mathbb{R}^{d_h} \to \mathbb{R}^{d_\phi}$ with d_ϕ=256 by *fixed seeded monomials*: the linear terms u_i plus a fixed, seeded selection of second-order cross-products $u_i u_j$,

$$\phi(u) \;=\; \big[\, u_1, \ldots, u_{d_h},\;\; u_{a_1} u_{b_1},\, u_{a_2} u_{b_2},\, \ldots,\, u_{a_m} u_{b_m} \,\big] \in \mathbb{R}^{d_\phi}, \tag{2}$$

where the index pairs $\{(a_r, b_r)\}_{r=1}^m$ are drawn *once* from a fixed seed and stored as constant integer buffers (here $m = 224$ cross-terms plus the 32 linear terms give d_ϕ=256). Crucially, ϕ has *no learnable parameters*: the monomial structure is a seeded constant, so QUAD2 adds **zero parameters** to the model. Parameter-matching to the Transformer is therefore exact-by-construction — the projection budget is unchanged.

2.3 Rectangular delta-state and why $O(1)$ inference survives

Replacing the keys k_t in Equation (1) with their lifted form $\phi(k_t) \in \mathbb{R}^{d_\phi}$ (and reading with $\phi(q_t)$) turns the carried state from a square $d_h \times d_h$ matrix into a **rectangular** $d_h \times d_\phi$ matrix:

$$S_t \;=\; \alpha_t\, S_{t-1} \;+\; \beta_t\, \varepsilon_t\, \phi(k_t)^\top, \qquad S_t \in \mathbb{R}^{d_h \times d_\phi}, \qquad o_t \;=\; S_{t-1}\, \phi(q_t). \tag{3}$$

This is the paper's contribution: the *rectangular delta-state framing*. The recall ceiling is governed by the effective rank of the lifted key space (d_ϕ near-orthogonal directions per head rather than d_h), recovering associative capacity, while the per-token write and read remain a pair of matrix-vector products over a state of *fixed* size $d_h d_\phi$ floats. Because that size does not depend on the sequence length n, the inference profile is unchanged from the linear case: per-step cost is $O(d_h d_\phi)$ and the carried state is constant in n. We verified that the single-step recurrence and the chunk-parallel training form agree ($\|\text{step} - \text{forward}\| < 10^{-6}$: none 3.6×10^{-7}, QUAD2 4.8×10^{-7}), so the cheap inference path is the exact same model as the trained one.

The cost of the lift is FLOPs, not parameters or memory: the d_ϕ=256 expansion multiplies the delta-state FLOPs and brings Prizma-Seq's forward cost to 2.14× the Transformer's per token at the headline scale (Section 7). We make no per-FLOP claim and disclose this ratio plainly.

3 Experimental setup

Pre-registered falsifiable bar. Before running the headline experiments we committed to a falsifiable bar consisting of the field-standard attention diagnostics plus a structural-advantage and a causal-attribution leg. The pass conditions (e.g. MQAR parity within noise at matched parameters; induction ≥0.98 with the 64–256 gap bin ≥0.95; selective-copy ≥0.97 with a fixed-position control; character LM test BPC $\leq T + 0.05$; constant-memory inference; and a causal control in which a random *linear* map matches the no-feature-map baseline) were fixed in advance. A leg can fail; the report states whatever the experiment shows.

Baseline and parameter-match auditor. The baseline is a tuned decoder-only Transformer with RMSNorm, SwiGLU, and RoPE, sharing the FFN, norms, embeddings, and tied head with Prizma-Seq so that only the mixer differs. A parameter-match auditor counts both models at each scale; the headline MQAR scale d128L4H4 is matched to **TF 857,216 vs. Prizma-quad2 862,368 params (+0.6%)**, and the character-LM scale to TF 3,220,480 vs. Prizma-quad2 3,224,608 (−0.13%). Scale notation is d{model}L{layers}H{heads}; throughout, d_h=32 and QUAD2 uses d_ϕ=256.

Protocol and honesty notes. MQAR and induction exhibit a seed-dependent phase transition at this scale *for both* architectures, so we report median, solve-rate (>0.9), and seed range over ≥3 seeds rather than a fragile mean. The MQAR/induction/selective-copy/causal legs ran at a single shared lr=10^{-3} (the Transformer's tuned value, which is *conservative for* Prizma-Seq, disadvantaging it, not the baseline); only the character LM swept a per-model learning rate. We disclose two reproducibility caveats binding on the diagnostic legs: per-seed weight init is not bit-pinned by the seed in those runs (the comparison is symmetric across architectures, but absolute solve-rates are not bit-reproducible), and the seed counts (n=3 diagnostics, n=2 character LM) are

Table 1: Prizma-Seq against the pre-registered, parameter-matched bar (tuned Transformer baseline, A100). Numbers are medians over seeds; see Sections 5–7 for caveats. Char-LM is reported as competitive *within* the +0.05 margin, *not* a win.

Leg	**Verdict**	**Headline**
MQAR (D=128)	PASS	Parity at 860K matched params (QUAD2 median 0.998 vs. TF 0.9995); solves at 130K where the matched TF fails (0/3) and the smallest grid TF solver needs $\geq$461K $\Rightarrow$ $\geq$3.5$\times$ param-efficiency (coarse grid).
Induction	PASS	QUAD2 median 0.9995 (3/3) vs. TF 0.9956. *Non-discriminating:* Prizma-none also solves (1.0).
Selective-copy	PASS	Selective median 0.9991 ($\geq T - 0.02$); fixed-position control isolates content-selectivity (spread 10^{-4}). *Non-discriminating:* Prizma-none also solves.
Char-LM (text8)	PASS	Prizma 1.7496 vs. TF 1.7254 BPC; margin $-0.024 <$ +0.05. Within the bar; does **not** beat the TF.
Inference	PASS (memory)	Constant 17.9 MB state $\forall n$ (28–455$\times$ less than KV-cache); measured $O(1)$-latency crossover at $n \geq$32k (2.4–2.8$\times$ faster at n=65k).
Causal ablation	PASS	QUAD2 (3/3, 0.997) $\gg$ rand_linear ($\approx$0.59) $\approx$ none ($\approx$0.52) $\gg$ TF (0.016). The gain is the monomials, not "a bigger RNN."
Length-extrapolation	WIN (relative)	Retention at 8$\times$ train length: Prizma 0.398 vs. RoPE-TF 0.041 ($\sim$10$\times$). Absolute acc still only $\sim$0.40 at 8$\times$.

descriptive, not powered equivalence tests. Inference *memory* is analytically counted from closed-form float counts; inference *latency* is measured wall-clock on an A100 (median of 5 reps, 2 warmup). All experiments are small scale on the named tasks — exactly what the bar pre-registered.

4 Results

Table 1 reports the seven legs of the pre-registered bar with headline numbers. All four field-standard diagnostic legs (MQAR, induction, selective-copy, character LM) pass parameter-matched against the tuned Transformer; the structural-advantage leg passes on memory at every length and on latency only at long context; the causal leg attributes the gain to the quadratic monomials; and length extrapolation is a *relative* win.

MQAR: parity and parameter-efficiency. At $\sim$860K matched parameters (d128L4H4, 3 seeds), all three arms solve D=128: Prizma-quad2 seeds $[0.9991, 0.9983, 0.9955]$ (median 0.9983) vs. Transformer $[0.9787, 0.9995, 0.9998]$ (median 0.9995). With n=3 at ceiling we cannot resolve a difference — this is descriptive parity (a failure-to-reject), not a tested equivalence. At matched params Prizma-quad2 (0.9983) also exceeds Prizma-none (median 0.9583), and ignites fastest of the three (by step 16–20k, vs. TF 44–80k). At the smaller 130K scale (d64L2H2), Prizma-quad2 solves D=128 (3/3, seeds $[0.999, 0.997, 0.935]$, median 0.997) where the param-matched Transformer fails (0/3, 0.016). The smallest Transformer that solves D=128 on our coarse 4-point grid $\{130, 461, 857, 3300\}$K is d128L2H4 (461K), giving a $\geq$3.5$\times$ parameter-efficiency gap. We flag that one quad2 seed (0.935) is borderline (clearing the 0.9 bar by 0.035) and that a denser Transformer

Table 2: Measured per-step decode latency (ms) and peak memory (MB) vs. sequence length n on an A100 (small config: d128L4H4, d_h=32, d_ϕ=256). Prizma-Seq latency is flat ($O(1)$); the Transformer grows. The latency crossover is at n=32k; below ∼16k Prizma-Seq is slower (overhead-bound).

n	4,096	8,192	16,384	32,768	65,536
TF latency (ms)	4.47	4.46	5.39	8.95	16.95
Prizma latency (ms)	6.96	6.95	6.95	6.95	7.06
TF peak mem (MB)	49.5	84.9	152.8	289.1	561.7
Prizma peak mem (MB)	17.9	17.9	17.9	17.9	17.9

sweep (200–450K) is needed to pin the true crossover.

Capacity question, resolved honestly. A sufficiently large Transformer *does* solve D=128 (d128L2H4/461K: 3/3, 0.9999; d256L4H8/3.3M: 3/3). The tiny-Transformer failure is under-capacity at that size, not "attention cannot recall." The honest differentiators are parameter-efficiency and $O(1)$-memory inference — never "attention fails."

Induction and selective-copy. Both pass parameter-matched (d128L2H4, 0.64% match): induction QUAD2 median 0.9995 (3/3, 0.9995 at the 256-gap bin) vs. TF 0.9956; selective-copy QUAD2 median 0.9991 vs. TF 0.9994, with the fixed-position control spread at 10^{-4}. We disclose that these two legs are *non-discriminating*: Prizma-none also solves them, so the feature-map's edge is visible in the MQAR-param-efficiency and causal-ablation legs, not here.

Character LM (text8). On text8 [12] (train 10M / val 0.5M / test 0.5M chars, V=27, AdamW [11, 10] wd=0.1, val-based early stopping, n=2 seeds), Prizma-quad2 reaches median test 1.7496 BPC vs. Transformer 1.7254 BPC — a margin of −0.024, within the pre-registered +0.05 bar and far below the random baseline of 4.7549. We state plainly: Prizma-Seq is competitive *within* the margin and does **not** beat the Transformer here. Neither arm overfits (final BPCs 1.76 / 1.74). The Prizma arm took ∼1889 s vs. the Transformer's ∼361 s (∼5.2× slower to train).

5 Inference efficiency

The structural advantage of a fixed-size carried state is constant inference memory. Decoding via `model.step()` (KV-cache for the Transformer, $O(1)$ state for Prizma-Seq) on an A100, Prizma-Seq's peak state is a **constant 17.9 MB at every sequence length** (small config, d_h=32, d_ϕ=256; 147,456 state floats), while the Transformer KV-cache grows linearly (49.5 MB at n=4k → 561.7 MB at n=65k). Analytically the KV/state ratio runs from 28.4× at n=4k to 455× at n=65k. The larger config shows the same pattern (Prizma 232 MB constant vs. TF 491 MB → 4532 MB; 19.5× at n=65k).

Table 2 reports measured per-step latency. Prizma-Seq's latency is statistically flat across a 16× span of n (∼7 ms throughout), while the Transformer grows from 4.47 ms to 16.95 ms. The wall-clock *latency* crossover is at n=32,768; at n=65k Prizma-Seq is 2.4× faster (small config: 7.06 vs. 16.95 ms) and 2.8× faster on the larger config (13.87 vs. 38.90 ms). **Below $n \lesssim$16k Prizma-Seq is ∼1.3–1.6× slower** (it is overhead-bound at short context): the latency win is long-context-only and we disclose it as such. The memory advantage, by contrast, holds at every length.

6 Ablation: is it the quadratic monomials?

To rule out the alternative explanation that the gain is merely "a bigger recurrent state" (a wider d_ϕ), we run a causal control at d64L2H2 (130K) on MQAR D=128 with three feature-map conditions plus the Transformer, all at fixed lr=2×10^{-3} (3 seeds):

$$\underbrace{\textsc{quad2}}_{3/3,\ \text{med } 0.997} \gg \underbrace{\text{rand\_linear}}_{0/3,\ \text{med } 0.589} \approx \underbrace{\text{none}}_{0/3,\ \text{med } 0.518} \gg \underbrace{\text{TF}}_{0/3,\ 0.016}.$$

The `rand_linear` control is a same-dimension random *linear* map (rank $\leq d_h$, so it expands d_ϕ to 256 without adding any nonlinear structure). It performs like the no-map baseline, not like QUAD2. This isolates the effect: the recall gain comes from the *quadratic monomials* specifically, not from any d_ϕ=256 widening of the state. Only QUAD2 solves D=128 at 130K parameters.

7 Limitations

We state every limit plainly; the project's credibility rests on not inflating the claim.

1. **Char-LM is within-margin, not a win.** On text8 Prizma-Seq is competitive *within* the pre-registered +0.05 bar (1.7496 vs. 1.7254 BPC) but does **not** beat the Transformer.
2. **Training is ∼5× slower per step.** The sequential delta recurrence makes Prizma-Seq ∼5× slower to train (char-LM: 1889 s vs. 361 s per arm).
3. **Latency win is long-context-only.** The measured per-step latency crossover is at $n \geq$32k; below ∼16k Prizma-Seq is ∼1.3–1.6× slower. The memory advantage holds at all lengths; the latency advantage does not.
4. **No per-FLOP claim.** Prizma-Seq costs 2.14× the Transformer's forward FLOPs per token at the headline scale. The two FLOP-matched Transformer arms (d128L9H4: 1/3; d208L4H4: 0/3, median 0.022) were optimization-confounded (consistent with a learning-rate-transfer artifact, since the same-depth narrower d128L4H4 solves), so they are excluded from the verdict and we make **no per-FLOP claim**. Equal-parameter parity already addresses "is it just compute?".
5. **Seeds are descriptive, not powered.** n=3 (diagnostics) / n=2 (char-LM) support failure-to-reject / descriptive parity, not a powered equivalence test. The diagnostic legs ran at a single shared lr=10^{-3} (conservative for Prizma-Seq) and their per-seed init is not bit-pinned by the seed.
6. **Length-extrapolation is relative.** Prizma-Seq retains 10× more accuracy than a RoPE Transformer at 8× train length, but its own absolute accuracy at 8× is only ∼0.40.
7. **No large-scale-LM parity and no backprop-free parity claim.** We test ≤3.2M parameters at character level; large-scale LM parity and backprop-free parity are open frontiers, explicitly not claimed.
8. **The kernel is borrowed.** QUAD2 belongs to the Based/Hedgehog feature-map family; the novelty is the *rectangular delta-state framing*, not the kernel.

8 Related work

Linear and feature-map attention. Linear attention [9, 3] reframes attention as a recurrence with a fixed-size state. Based and Hedgehog [2, 23] restore the recall lost by linearization through expressive feature maps, including quadratic ones; we credit the QUAD2 kernel to this family. Our contribution relative to them is the *rectangular* ($d_h \times d_\phi$) delta-state framing and its parameter-free, fixed-seeded-buffer instantiation.

Delta-rule and gated recurrences. The delta rule connects to fast-weight programmers [15]; DeltaNet [21] parallelizes it over sequence length, and Gated DeltaNet [20] adds a data-dependent gate, as in Mamba [7]. Prizma-Seq's mixer at the parallelizable level coincides with Gated-DeltaNet plus a short conv and a local window head — all borrowed, known-good components; we make this explicit rather than presenting a new primitive.

State-space models. S4 [8], Mamba [7], and the SSM-attention duality [4] establish constant-state sequence mixing; Griffin [5] combines gated recurrences with local attention, motivating our window head.

Recall diagnostics. MQAR and the Zoology suite [1] are the field-standard probes for the recall–efficiency tradeoff; our pre-registered bar uses them directly.

Predictive coding. The free-energy reading of the delta write draws on predictive coding [14, 6]; the optional local/random-feedback training mode relates to direct feedback alignment [13] and is a secondary axis, not a gate here.

9 Conclusion

Prizma-Seq is a Gated-DeltaNet–family sequence mixer whose only lever is a parameter-free quadratic feature map (QUAD2) that makes the carried delta-state rectangular ($d_h \times d_\phi$) at zero added parameters and with $O(1)$ inference preserved. Parameter-matched against a tuned Transformer it clears a pre-registered diagnostic bar at small scale: descriptive MQAR parity at 860K params, $\geq$3.5× parameter-efficiency at 130K, solved induction and selective-copy, a within-margin character LM, constant-memory inference with a long-context latency crossover, and a causal ablation attributing the gain to the quadratic monomials. It is a *candidate* efficient-attention replacement in the tested regime — not a proven alternative. The open frontiers we do not claim are large-scale LM parity, per-FLOP advantage, and backprop-free parity.

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